FINAL-YEAR PROJECT · VIVA / DEFENSE PREPARATION

Defending the Project

Everything you need to walk into the room ready: a one-minute pitch, a clean spoken walkthrough of the system, the numbers that matter committed to memory, and honest, panel-tested answers to the questions you will actually be asked — including the hard ones.

The 60-second opening

If you say nothing else, say this. It frames the project on your strongest ground before any question is asked.

> "The problem I set out to solve is that a language model can't reliably tell you whether its own answer is correct — and in my domain there's **no answer key to check against**. So I built a pipeline where two models generate answers, two more evaluate them blindly on a shared rubric, and three external checks corroborate the result, all combined into one confidence score that decides whether to ship the answer or try again.
>
> The hard part wasn't building it — it was **proving each piece works without any ground-truth labels**. Every model choice, every parameter, is backed by a label-free experiment. And along the way I caught three of my own measurement errors before they became conclusions — which is the real reason I trust the results I'm about to show you."

The spoken walkthrough

Six steps, one sentence each. Practise until it's smooth — this is your "explain your system" answer.

1. **Two models generate, independently**
   LLaMA 3 and Mistral each answer the question from scratch — two independent shots, not one.

2. **A verifier scores both, blind**
   phi3 rates each answer on eight rubric criteria — factual correctness, faithfulness, hallucination-freedom, and so on — without knowing which model wrote it.

3. **A separate judge scores again and picks a winner**
   qwen3.5:9b does the same eight-factor scoring completely independently, then declares a winner — the two independent evaluations let me measure whether they agree.

4. **Three external checks corroborate**
   Cross-model agreement, plus live Wikipedia and Wikidata lookups, both gated so an irrelevant source is never trusted.

5. **Six signals become one confidence score**
   A weighted blend, no single signal dominating, produces a 0–1 confidence — and if it's below 0.72, the whole thing regenerates.

6. **A combiner writes the final answer**
   The best attempt is synthesised into the answer that ships — and every raw signal from every attempt is logged for analysis.

Numbers to have memorised

Panels trust candidates who quote their own figures without hunting for them. These are the ones worth knowing cold.

5,000
questions, one consistent regime

8,432
total attempts logged

0.72
confidence threshold, CI [0.69, 0.75]

0%
judge flat-rate (vs 57.5% for the rejected model)

55.3%
Mistral win-rate by true identity

74.5
verifier discrimination gap /80 (solid vs broken)

32/37
rows exposing the judge-copying bug

p=0.011
significance of the threshold finding

Anticipated questions & answers

Ordered from most likely to hardest. Each answer leads with the one-line version, then the backup detail if they press.

FOUNDATIONAL · VERY LIKELY

Q1 Why should anyone trust an LLM to judge another LLM?

Lead: "Because I never assumed it — I tested it." **Backup:** I ran a discrimination test — a deliberately broken answer against a solid one. A judge that can't score the broken one far lower is unusable. My final judge scored solid answers 77–80 out of 80 and broken ones 0–8. I also rejected a judge that had good average scores but gave all eight rubric criteria the identical value on 57.5% of answers — a fluent number with no analysis behind it. Trust came from the test, not the fluency.

METHODOLOGY · VERY LIKELY

Q2 You have no ground-truth labels. So how do you validate anything?

Lead: "With label-free objectives that are still falsifiable." **Backup:** Two families. First, discrimination — can an evaluator tell bad from good on constructed cases. Second, internal consistency and regeneration validity — do the signals predict each other, and does a low score genuinely predict that a retry improves the answer. Neither needs a correctness label, but both can fail, which is what makes them real tests rather than assumptions.

> **If they push:** concede openly — "This measures self-consistency and corroboration, not verified correctness. Real accuracy would need a human-labelled subset, which is exactly my proposed next step." Owning the limit is stronger than defending past it.

RIGOR · LIKELY

Q3 How do you know your scores measure quality and not something else?

Lead: "I found out they partly didn't — and fixed it." **Backup:** I discovered LLaMA 3 was always shown as 'Answer 1' and the prompts named the brands, so position and identity were confounded with quality across the entire early dataset. I made evaluation blind and randomised answer order per question. Once separated, the raw-position win gap of 35/65 shrank to a true-identity gap of 45/55 — proving most of the apparent gap was an artefact, and quantifying the small part that was real.

STATISTICS · LIKELY

Q4 Why 0.72 for the threshold? Isn't that arbitrary?

Lead: "Three independent methods agree on it." **Backup:** The problem was censored data — you only ever see retries below the threshold. I added a control group: a random fraction of already-passing answers get a retry too, purely to measure them. Then a z-test (p=0.011), an ROC/Youden's-J analysis, and a 2,000-sample bootstrap all landed on a 0.69–0.75 optimal range. I adopted 0.72 — the bootstrap-median optimum — which sits right in the middle of that plateau, not a knife-edge guess.

THE HARD ONE · BE READY

Q5 Your system says Mistral is better. Is it, really?

Lead: "Honestly — I can't fully prove that, and here's exactly why." **Backup:** The judge rates Mistral higher on factual criteria. But the independent verifier sees no difference at all on the same answers — and I proved the verifier isn't just blind, because it discriminates broken answers cleanly. So the balance of evidence says the judge's preference may be a model-specific perception, not a real quality gap. Settling it would need per-answer fact-checking I don't have at scale. I report it as open rather than overclaiming.

> **Why this is a strength:** a panel trusts a candidate who says "the evidence doesn't settle this" far more than one who forces a clean story. This answer demonstrates you understand your own results' limits.

INTEGRITY · IMPRESSIVE IF IT COMES UP

Q6 Did anything go wrong during the project?

Lead: "Yes — three measurement bugs, all caught before they became conclusions." **Backup:** One, a model reported a total that didn't match its own criterion scores — I recompute totals now, never trust the model's arithmetic. Two, a 'second-opinion' signal was secretly derived from the judge itself, so it inflated results — I excluded it from the search. Three, a threshold analysis under-counted part of the data because a sampling rate had changed mid-run — I re-weighted by the rate actually in force. Each was found by a result looking suspiciously clean and being checked, not by luck.

SCOPE · LIKELY

Q7 Isn't this just a data-generation pipeline? Where's the "intelligence"?

Lead: "The intelligence is the self-validation, not the generation." **Backup:** Any pipeline can produce answers. The contribution is a system that decides how much to trust its own output, corroborates it, and retries when unsure — and a demonstrated method for proving each of those decisions is sound without a labelled dataset. The 5,000-question corpus is the evidence that the method works at scale, and the raw material for a follow-on confidence-predictor model.

GENERALISATION · POSSIBLE

Q8 Would this work with different models, or in a different domain?

Lead: "The method transfers; the specific numbers wouldn't." **Backup:** The protocol — blind evaluation, discrimination testing, control-group calibration — is model- and domain-agnostic. What's specific to my setup is the tuned values: the exact threshold, the weights, which judge won. On a new model set you'd re-run the same experiments and get new values. I haven't tested that transfer, and I'd state it as a limitation rather than claim it.

Delivery notes

Small things that change how the room reads you.

- **Lead with the honest limits before they ask.** Naming what your evidence doesn't cover — accuracy, the Mistral question, generalisation — makes every claim you *do* make more credible.
- **Turn the bugs into the highlight.** The three self-caught errors are your strongest material; a project with zero visible mistakes reads as one that didn't look hard enough.
- **Quote a number, don't reach for notes.** "57.5% versus 0% flat-rate," "p equals 0.011," "gap of 74.5 out of 80" — specifics signal command of your own work.
- **If you don't know, say what you'd do to find out.** "I haven't tested that — the way I'd check it is..." is a complete, confident answer.
- **Have the repo and the live demo ready.** Being able to show the pipeline running, or open the dataset viewer, converts a claim into a demonstration.